Network Scanner

Problem statement and background:

Given a set of network traffic logs with features describing connection metadata and behavior, predict whether each connection instance represents a normal (benign) event or a malicious (attack) event — and ideally identify the type of attack.

With the exponential growth of internet-connected systems, cyberattacks have become more sophisticated and frequent. Network administrators need automated and intelligent systems to detect and prevent malicious activity in real time. Traditional intrusion detection systems (IDS) rely heavily on static rules and signatures, which cannot keep up with new or evolving attack patterns.

Machine learning offers a dynamic alternative: by training models on labeled network traffic data, we can detect patterns that distinguish normal traffic from various types of attacks.

The CIC-UNSW-NB15 dataset, created by the University of New South Wales and the Canadian Institute for Cybersecurity, provides a realistic and modern dataset for evaluating intrusion detection systems.

Data Wrangling:

The raw dataset from the Canadian Institute for Cybersecurity has 447,915 rows and 83 columns. This is too much data to try and process all at once and out of scope for this project. Some of the data includes redundant pieces of data, but good news there is no missing data from this set. The zeros in the data are just as important as the numeric data. There will be ten different types of network scenarios that I will be testing for.

These are as follows, Benign or normal activity, Analysis or gathering information about the network, Backdoor or ways that attackers will maintain access to the

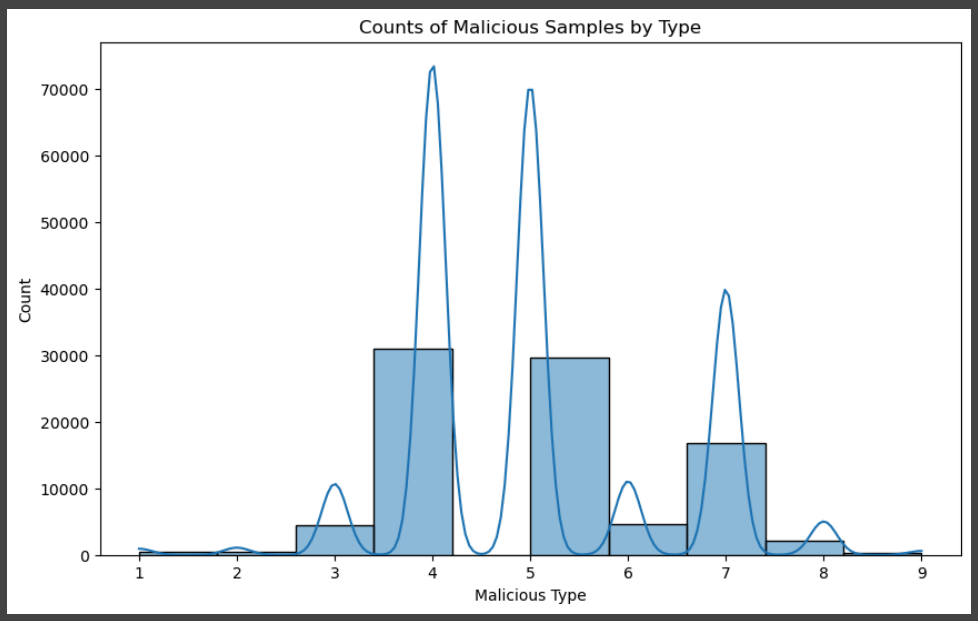
network, Denial of Service Attacks also known as DoS (this attack floods the network with transaction slowing or completely overwhelming the network causing the system to stop work as intended), and Exploits is a huge category due to the diversity of types of attacks, We will cover this a little more in depth later. For the remaining types of attacks, Fuzzers, Generic, Reconnaissance, Shellcode, and Worms.

The types of Attacks:

- **Analysis:** This isn't really an attack most of the time, but could lead to one. When looking at analysis, or network scans, most attackers are looking for vectors that are able to attack. Scanning Ports, Sniffing traffic, or overall just monitoring the system.
- **Backdoor:** This happens after an attack normally, someone got in and left themselves a door to return to.
- **DoS:** This attack floods the network with transaction slowing or completely overwhelming the network causing the system to stop work as intended
- **Exploits:** These are hardware dependent attacks, meaning this attack only works with X device with Y version of firmware. And the same type of attack won't work on X device with Z version of firmware.
- **Fuzzers:** An automated attack technique that floods systems with malformed, unexpected, or random data packets to expose security vulnerabilities, crashes, or memory leaks.
- **Generic:** Unauthorized attempts to disrupt, steal data from, or gain access to IT infrastructure. Example: Employee plugs an unidentified usb into their work PC, this usb has malware on it and now the attack was successful.
- **Reconnaissance:** These are active scans to the network usually more targeted than Analysis.
- **Shellcode:** An Attacker has found a way to execute shell code to the system gaining control. Rare but can happen.
- **Worms:** A type of self-replicating malware that spreads autonomously across networks by exploiting software vulnerabilities, without requiring user interaction

EDA:

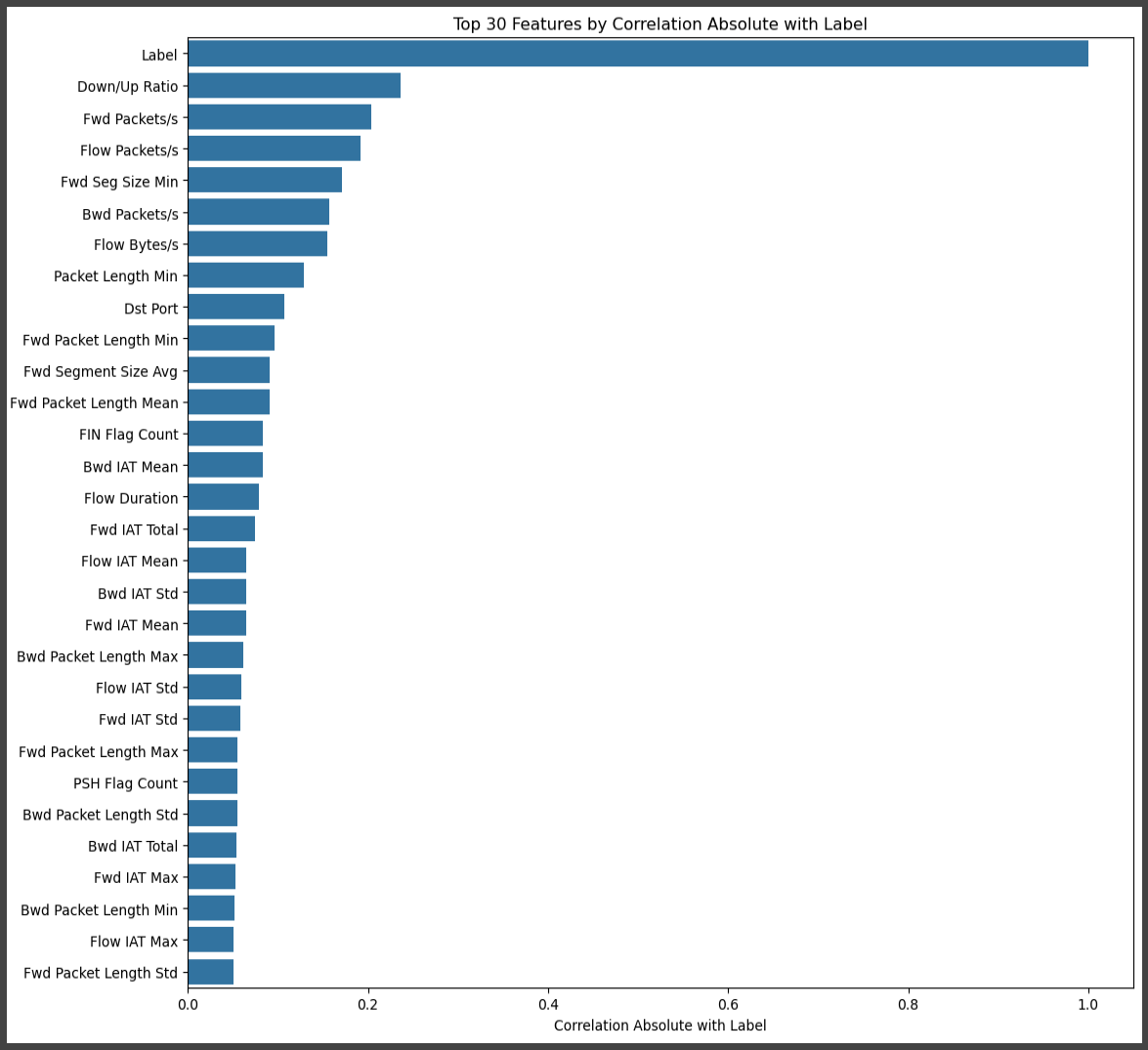
Taking a closer look at the data, there are several columns that have little to no variety. Also, this data set is too big to realistically model and process for the computing power I have access to. So sampling the data set well maintains the distribution of the data set (below, distribution of attack types without benign data due to the huge skew of the data).



After the sampling 100,000 is done the distribution of Attack types look like:

Benign	97,426
Analysis	11
Backdoor	12
DoS	128
Exploits	889
Fuzzers	850
Generic	133
Reconnaissance	480
Shellcode	60
Worms	7

Now let's talk about the correlation between each parameter compared to the label. Correlation is a percentage that determines how much one parameter affects the outcome in this cause the label or attack type. In the graph below, we can determine that there is no real strong connection between the network data, and an attack type. Meaning there is no single param we can test to determine if we are under attack on the network or not. Good news for us is that we can use multiple parameters to detect an attack. Another way we will try to detect an ongoing attack is through outliers, but to network activity varying between task to task we might get some false positives when detecting threats.



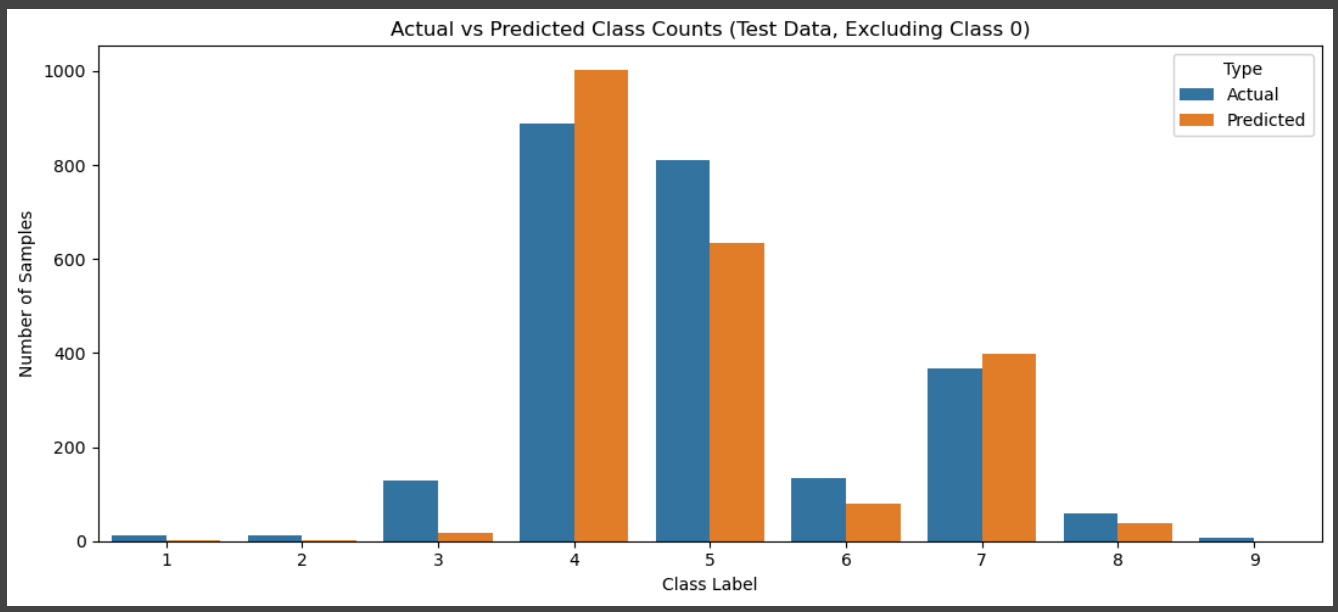
These 30 params are the ones we well use to build our model due to having the highest impact on the label. Now taking a closer look at the top 10, lets see if maybe we can find some unique characteristics for the attack types. Notable is that the DoS typically shows high packet rates and connection counts. Where exploits follow a pattern of specific protocol or packet size signatures. Backdoors have longer persistent connection patterns. Benign data has low flow packets, low amounts of forward packets, and also uses a reliably one port.

The Model:

The goal of this project was to create a model capable of detecting threats and what kind of threats they are. While modeling I chose a decision tree approach to processing the data. I compared XGBoost, Random Forest, Gradient Boosting, Logistic Regression, and LightGBM. I used a grid search method to try each of these models, also finding the best params for each model. Here are some of the findings:

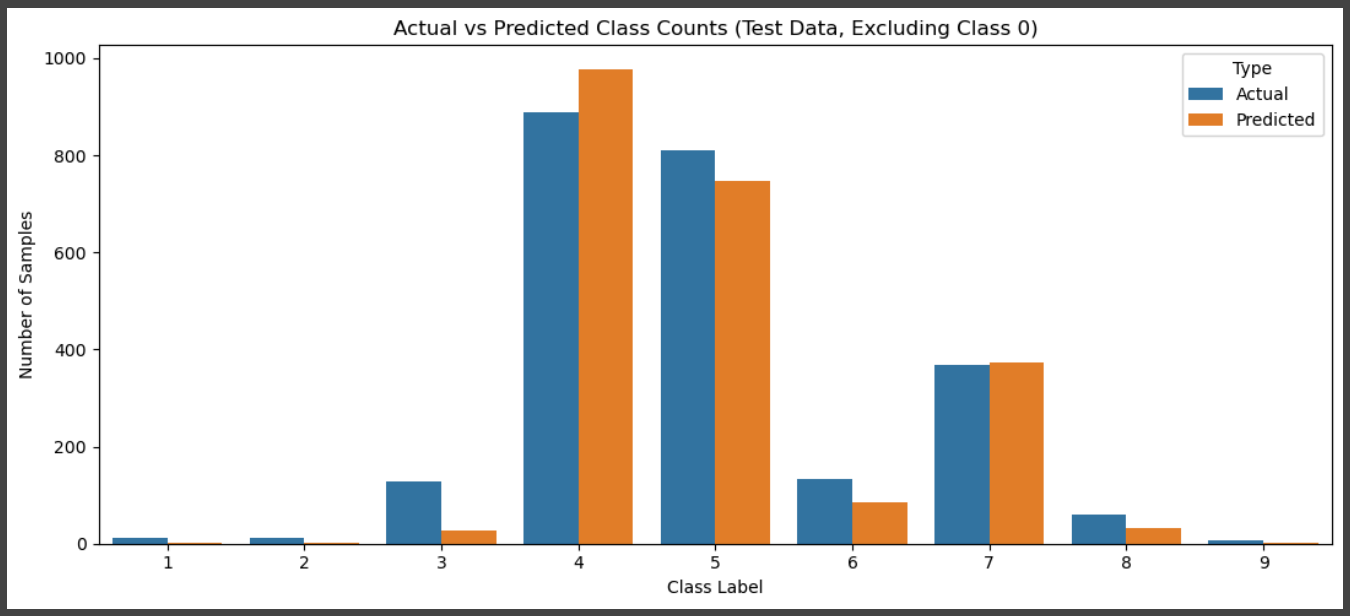
Model	Accuracy Score
XGBoost	0.9856
Random Forest	0.9849
Gradient Boosting	0.9829
Logistic Regression	0.9664
LightGBM	0.9859

Just going off of accuracy scores isn't enough to pick the best model due to XGBoost and LightGBM being so close. So let's take a closer look at the actual prediction by the model compared to the actual answer.
First XGBoost:



It over-predicted on class 4 which is Exploits, and really under performed on class 3 (DoS).

Second LightGBM:



LightGBM was very similar to XGBoost, but with a few key differences, the over prediction isn't as high and the under predicted categories on the LightGBM is closer to the actual than the GXBoost. Notice 7 and 5 for example, Reconnaissance and Exploits, 7 is much closer to the actual than in the graph from XGBoost.

One down side to trying to predict a threat with network data, is the overwhelming variance in the data, just simply in the benign data for example, just a slow connection can make it seem like a threat like when in reality it's just a slow connection. So no model I make will be 100% accurate, but we can get close.

So without any more delay, the model I chose to keep was the LightGBM model. Due to some key values, the accuracy score of 98.59%, and the accuracy at predicting benign data, at 99%. I would rather my model predict a threat as any threat other than benign because knowing there is a threat rather than assuming its benign can save the day even if the software reported the wrong threat. So I'm happy to say my model can predict a threat within 98% accuracy.

Citation:

H. Mohammadian, A. H. Lashkari, A. Ghorbani. “Poisoning and Evasion: Deep Learning-Based NIDS under Adversarial Attacks,” 21st Annual International Conference on Privacy, Security and Trust (PST), 2024.